



3D microwave printing temperature control of continuous carbon fiber reinforced composites

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ARTICLE INFO

Keywords:

Microwave processing
3D printing
Carbon fibres
Polymer-matrix composites (PMCs)

ABSTRACT

Continuous carbon fibers show dramatic promise as reinforcement materials to improve the stiffness, strength properties and design ability of 3D printed polymer parts. Current 3D printing methods have a low printing speed because the intrinsic slow and contact needed heat transfer disadvantages of the traditional resistive heating approach. We present a 3D microwave printing method by using the microwave for instantaneous and volumetric heating the continuous carbon fiber reinforced polymer (CCFRP) filament. This allows fabricating CCFRP components with much higher speed compared to traditional 3D printing technologies. To utilize the benefit of high printing speed, the speed-variation 3D microwave printing is applied to adapt the diverse printing path and reduce the printing period. In this paper, a new 3D microwave printing temperature control method by combining the prediction-model and step-proportional-integral-derivative control is researched to reduce the printing temperature difference of the CCFRP filaments during the speed-variation printing process. Three different CCFRP specimens with variation printing speed are tested, including a spanner, an aircraft and a spider from Nazca lines. The experimental results indicate that the new printing temperature control method for 3D microwave printing process dramatically reduces the temperature deviation. Further mechanical testing results indicate that the CCFRP printed with this method has a high tensile strength up to 358 MPa. This technology solved a key problem of 3D microwave printing of continuous carbon fiber reinforced polymer composites and can be used to manufacture complex polymer-matrix composite materials.

1. Introduction

Three-dimensional (3D) printing, also known as additive manufacturing, has been studied for more than 20 years for potential application in aerospace [1], automotive and medical treatment [2]. Benefit from the computer-aided design, numerical and automatic control, the 3D printing becomes a burgeoning method for manufacturing of composite materials and already leads to a revolution in the manufacturing industry.

The mostly used printing technologies for composite materials include direct ink writing [3], fused filament fabrication (FFF) [4–6], stereolithographic printing [7] and laminated object manufacturing [8]. Since 2016, the authors [9] researched the continuous carbon fiber reinforced thermoplastic composite printing by in-nozzle impregnation of FFF printing process and indicated that by introducing the continuous carbon fiber, the extrusion process turns into pultrusion. Matsuzaki et al. [10] mixed the continuous carbon fiber and polylactic resin to fabricate fiber reinforced composites which showed mechanical properties much

better than the unreinforced thermoplastics. Caminero et al. [11] demonstrated that the impact strength and interlaminar shear strength exhibited by 3D printed continuous carbon fiber reinforced polyamide composites were higher than the usual 3D printed thermoplastics. Tian et al. [12] showed that the mechanical properties of recycled carbon fiber filaments was higher than that of originally printed composites. By using the infrared heating, the surface temperature of the printed layer are increased and can improve the interlayer strength of the printed parts [13]. The authors also developed a path planning method to rearrange the tensile and compressive stresses and printed the topological optimized CCFRP [14]. Although the continuous reinforcement enhances the mechanical property of 3D printed plastics, but the printing speed of CCFRP still remains a big challenge, which inhibits its further industrial scale production. Ye et al. reported a speed of 5 mm/s to print the continuous carbon fiber reinforced thermoplastic polyimide [15]. The company Markforged published the Mark series 3D printer and its printing speed during fabricating continuous fiber reinforced polymer parts is about 7 mm/s [16], which is so far the highest speed

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reported.

These above research works have shared a common feature to employ traditional resistive heater to heat the filament. During the printing process, thermal heat transfers from heating block to the fiber, but relies deeply on the resin matrix which has a low heat conductivity coefficient. Therefore, the real situation is that the skin area of the filament already reaches above degradation temperature, while the core area is still below melting temperature, especially for high printing speed and large diameter filaments. One feasible method to solve this problem is to increase the contacting area between the resin and the heater (same mechanism of the screw extruder). However, this leads to high friction during the printing process and will easily break the continuous fiber. Besides the traditional resistance heating, the laser heating assisted laminated object manufacturing [17] and ultraviolet printing [18] of the carbon fiber reinforced plastic filament have been proposed, but still cannot penetrate and volumetrically heat the filament. As a result, the current 3D printing method can only manufacture small end-use products with extremely low printing speed and efficiency.

To deal with such problems that the contact heat transfer caused slow printing speed and non-uniform heating, we proposed the 3D microwave printing (3DMP) method of continuous carbon fiber reinforced thermoplastic composites with higher printing speed and performance. The first 3D microwave printer which was named as SERPENS (Super Efficient and Rapid Printing by Electromagnetic-heating Necessitated System) has already been developed by the authors. Based on this 3D microwave printing prototype, the microwave heating temperature control methods under variation printing speed are researched in this paper. Three different specimens with complex geometric characters, namely a spanner, an aircraft and a spider from Nazca lines, are printed by using different temperature control methods. The spanner has long straight printing path, the aircraft specimen contains sharp corners and lots of complex curves exist in the spider specimen. The optical structure images and mechanical properties are analyzed to further inspect the performance of the 3D microwave printed CCFRP parts.

2. Materials and methods

2.1. Materials used and the 3D microwave printer SERPENS

The 1K 66tex HT T300 carbon fiber (Toray Industries) and 910 nylon material (Taulam3D, melting temperature: 210 °C) were employed in this paper. The PA845H-PL nylon sizing agent (Michelman company) was used for the surface modification of the continuous carbon fiber bundle. The self-developed continuous carbon fiber impregnation system and 3D microwave printer SERPENS are applied, as shown in Fig. S1 of the supplementary information. The manufactured CCFRP filament has 0.45 mm diameter and about 25% carbon fiber volume content. A FLIR A35 infrared thermal camera was applied to measure the microwave heating temperature of the CCFRP filament inside the resonant applicator. A universal electronic testing machine INSTRON 4045 was used to test the specimens fabricated by obeying the testing standard of DIN EN ISO 527-2. A OLYMPUS BX60M microscope was employed to investigate the internal quality of the printed specimens.

2.2. Investigation of the 3DMP temperature control methods

In principle, the continuous carbon fibers are pulled out from the microwave applicator and adhere to the printing bed continuously [9]. If the printing path has a changing direction vector (such as a curve or turning corner), a persistent bonding force from the previous printed filament section is needed. Thus, the straight printing path has the direction vectors in one direction and can achieve the highest printing speed. But the curves or turning corners should apply a low speed to make sure the filament can bond to the printing bed. For the variation printing speed, an efficient microwave heating temperature control

method should be developed.

The temperature control strategy can be divided into two stages, the first stage is to heat the filament from room temperature to the melting temperature, and the second stage is to maintain the melting temperature under different printing speed. At the first stage (mark as stage 1 in Fig. 1), general proportional-integral-derivative (PID) control will be employed to adjust the microwave power and increase the filament temperature, as shown in Fig. 1. The main challenge of the 3DMP temperature control is the second stage, which needs to control the microwave power corresponding to the printing speed. The different temperature control methods for the second stage (mark as stage 2 in Fig. 1) are investigated, including the general PID, fixed microwave power, step-PID (output step-function signals) and the combination of prediction-model and step-PID control methods, as illustrated in Fig. 1.

The PID control has temperature feedback from the infrared camera and controls the microwave output power by pulse-width modulation (PWM), as shown in Fig. 1 (a). By using the PWM control signal, the microwave power generated from the solid-state microwave amplifier can be precisely adjusted with 0.1 W resolution continuously. The fixed microwave power control provides a constant microwave power during the printing process and does not relate with the variation of printing speed, as shown in Fig. 1 (b). By modifying the PID controller, step-PID is developed with a step-function control of the output signal. That means the microwave power will remain constant if the measured printing temperature has a slight fluctuation ($\pm 10^\circ\text{C}$). We presented the printing temperature control method by using the prediction-model to decrease the temperature difference during the 3D microwave printing process. To establish the prediction-model, the microwave heating power of the CCFRP filament was measured under difference printing speed. Then, the prediction-model based controller can quickly adjust the microwave power to the appropriate value when the printing speed changes (detailed information is described in the supplementary information). Due to the random communication delay (about 0.2 s–0.25 s) between the MatLab control code and the microwave amplifier, sometimes the microwave power control could not correspond to the printing speed (about 0.02 s–0.05 s leading or lagging). As a result, the step-PID is introduced to work with the prediction-model based controller to reduce the influence of random communication delay, as shown in Fig. 1 (c).

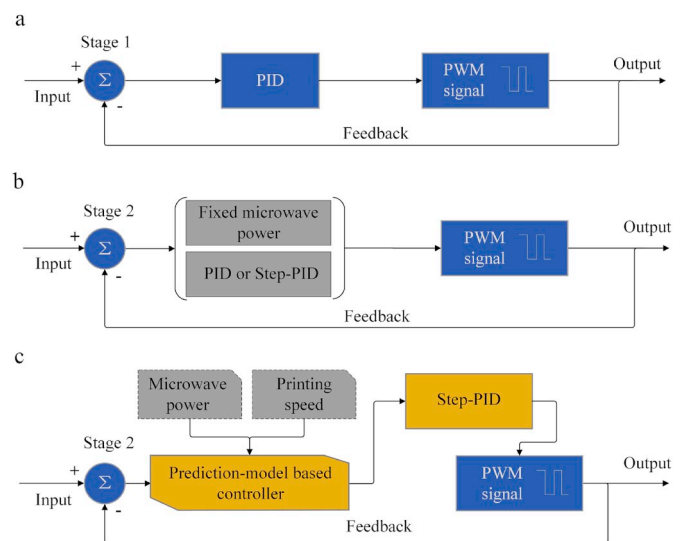


Fig. 1. 3D microwave printing temperature control methods of CCFRP, (a) PID control method for initial heating of stage one, (b) fixed microwave power and PID controlled printing process of stage two, (c) prediction-model and step-PID (PMSP) control method of stage two.

3. Results and discussion

3.1. 3D microwave printing temperature control of CCFRP

The microwave heating power of continuous carbon fiber reinforced nylon filaments (manufactured by the impregnation system) are measured under different printing speed, as shown in Fig. 2 (a). The relationship between the microwave power and printing speed has been measured for several times and the fitted data has been calculated. The fitted data are similar to a linear line and can be expressed as: $P = 10.5 + 0.5 \times V$. Where P is microwave power (W) and V is printing speed (mm/s). For a motionless filament in the microwave applicator, only about 10.5 W microwave power is needed to heat the filament to 210 °C. The fluctuation of the measured microwave power may be caused by the varying position of the 0.45 mm diameter filament in the 2 mm diameter nozzle tube.

The heating thermal image of the CCFRP filament is measured from a sight window (5 mm diameter hole on the applicator) and illustrated in Fig. 2 (b). Due to the infrared thermal camera measures the temperature of the whole area, no matter the filament has a position switching or moving, the microwave heating temperature can be captured precisely. The testing device is shown in Fig. S5 in the supplementary information. Before the printing process, the G-code will be revised to insert the command to output the speed control signal. The control signal will be adjusted according to printing speed, as shown in Fig. 2 (a).

To validate the efficiency of the 3D microwave printing temperature control method, three different specimens with variation printing speed is printed, including a spanner, an aircraft and a spider from Nazca lines (a biomorphic graph in the shape of a spider) [19]. These specimens have different geometrical characters to verify the printing temperature control method. The first spanner specimen has speed changing from the long straight lines to the curved sharp corners. In the middle of the two corners are the semicircles (about 4 mm diameter). The aircraft contains lots of short straight lines and the connection areas, which have the angles range from 30° to 87°. For the specimen of the spider shape, the complex curved paths are connected continuously. The three specimens represent the different printing paths that may happened usually during the printing of continuous carbon fiber reinforced composite parts. The printing path with variation printing speed was generated by calculating the radius of curvatures of curves and the angles between direction vectors of the path. The code was programmed in Matlab and the algorithm information is described in the supplementary information (Table S1).

As illustrated in Fig. 3 (a), the radius of curvatures of the printing path are calculated (the calculated angles between the adjacent

direction vectors are informed in Fig. S2 in the supplementary information). The different color point out the magnitude of the direction vectors and the arrows indicate the changing direction. After that, the printing speed diagram is achieved, the orange areas (straight and large curvature areas) have a printing speed of 20 mm/s, and the blue corner areas are 7 mm/s, as shown in Fig. 3 (b). The measured printing temperature and microwave power of different printing temperature control methods are exhibited from Fig. 3(d)–(h). It can be seen that the microwave has a very fast heating rate of the filament, and before the start of the printing process, a dwell stage at 210 °C is applied.

Obviously, the highest temperature difference of the CCFRP filament occurred when the fixed microwave power is imposed during the printing process. That is the first key difference between the traditional resistive heating 3D printing and the 3D microwave printing technology, constant heating power is failed for 3DMP. The filament temperature is very sensitive to the microwave power inputted in the applicator, due to the strong microwave absorption properties of continuous carbon fiber. Fig. 3(e) and (f) exhibit the curves of printing temperature and power under general PID and step-PID control methods. A similar standard deviation (in short of SD in the figure and indicates the dispersion degree of the printing temperature) of the printing temperature is measured of these two control methods, which is 44.3 °C and 47.6 °C respectively. Finally, the printed CCFRP spanner by using PMSP control method is shown in Fig. 3 (c). By comparing the printing temperature and the printing path, we found that the highest temperature difference occurs on the notched corner of the spanner, which is the speed changing areas.

Even if a small standard deviation is obtained compared with the fixed power control method, a maximum temperature value during the step-PID printing process can be discovered, as shown in Fig. 3 (i). Because, no matter how fast the temperature feedback is of the PID controller, it always delays to the changing of the printing speed, a strong regulation signal will generate and may cause overshoot of temperature. This is the second key difference between the traditional 3D printing and the 3D microwave printing technology, ex-post PID control methods (the widely used heating method in traditional 3D printing) lack of control ability of the 3DMP. Therefore, the new prediction-model based temperature control method is researched and shown in Fig. 3 (g) and (h). A new 3D microwave printing temperature control method is implemented and the measured data show small temperature variation and standard deviation, as illustrated in Fig. 3 (i). For this CCFRP spanner, 8 high-speed and 7 low-speed paths are calculated, which correspond to 8 peaks and 7 troughs of the square wave shown in Fig. 3 (g). The standard deviation of prediction-model and step-PID (PMSP) control method reduces by 69% and 26% of the step-PID and prediction-model control method, respectively. The PMSP

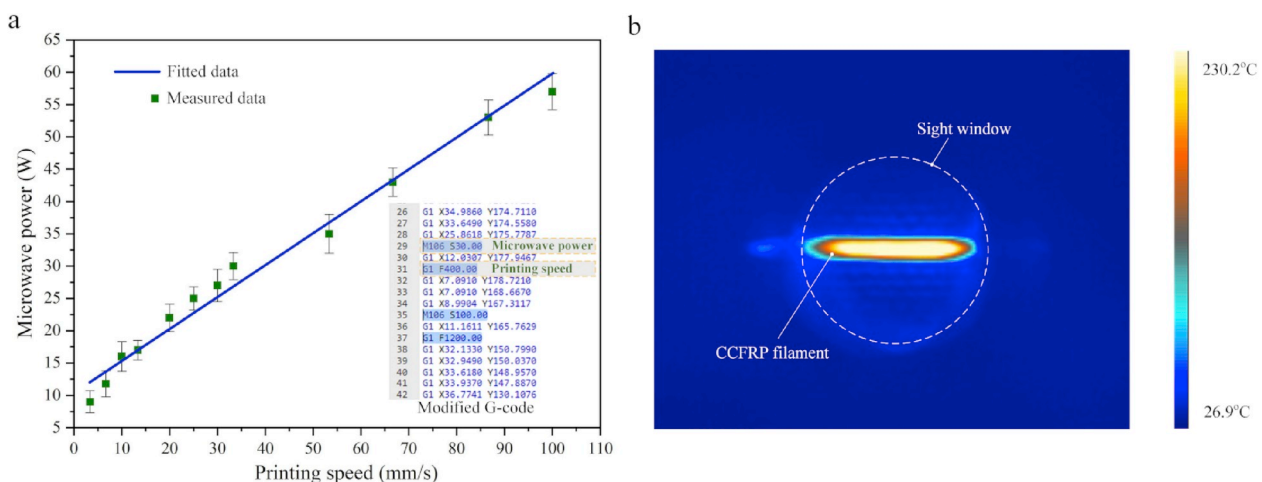


Fig. 2. Relationship between microwave power and printing speed, (a) measured microwave power at different printing speed and linear fit, (b) the measured thermal image of the heated CCFRP filament in the applicator.

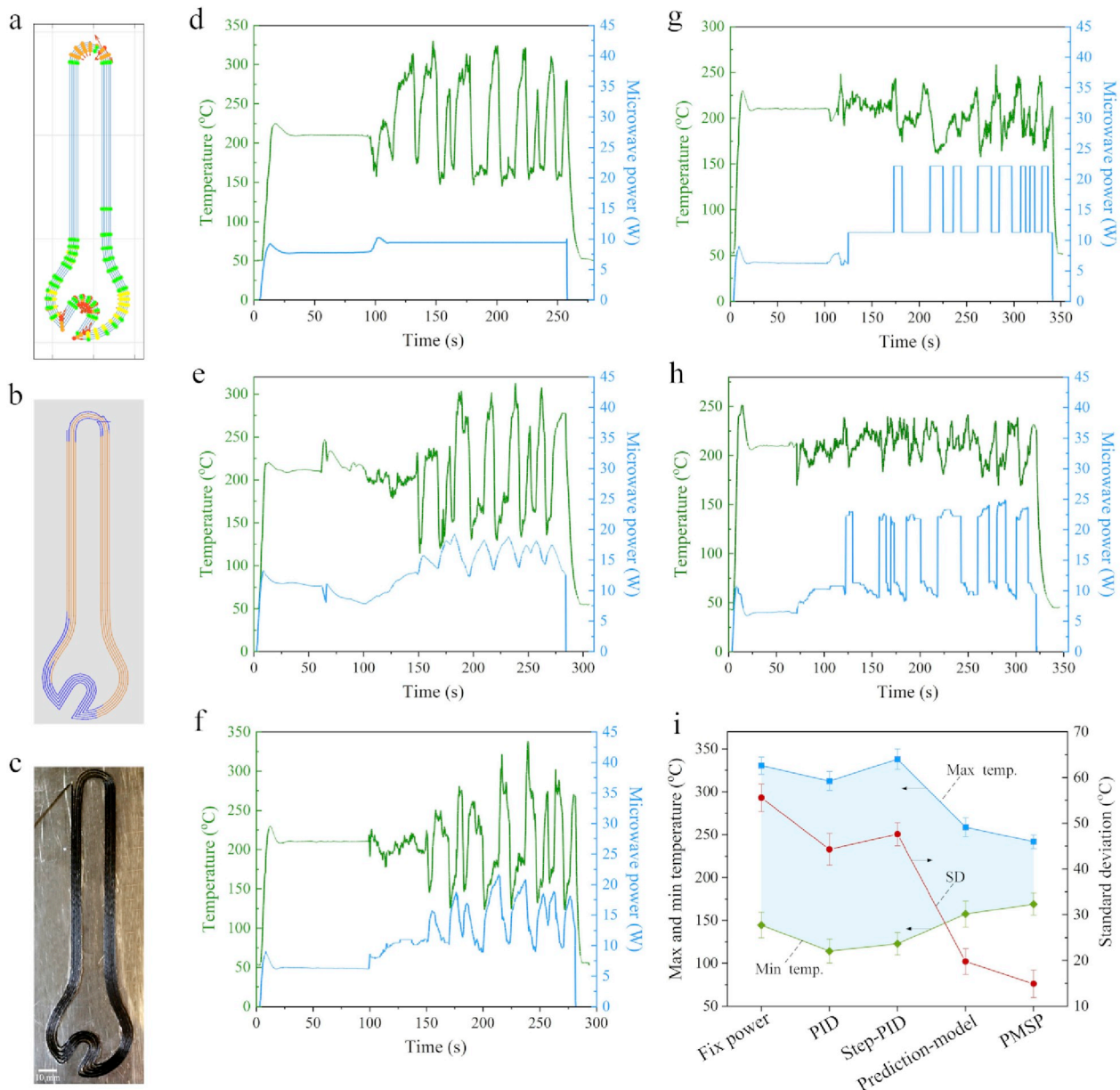


Fig. 3. 3D microwave printed CCFRP spanner and measured printing temperature during the printing process, (a) calculated radius of curvatures of the printing path, (b) printing speed variation along the printing path of the spanner, (c) the printed CCFRP spanner by PMSP method, (d) printing temperature under fixed microwave power, (e) printing temperature under general PID controlled microwave power, (f) printing temperature under step-PID controlled microwave power, (g) printing temperature under prediction-model controlled microwave power, (h) printing temperature under prediction-model and step-PID controlled microwave power, (i) the comparison of five different printing temperature control methods.

method dramatically reduces the standard deviation of printing temperature by 73% and 66% of the fixed power and general PID method, respectively.

In Fig. 4, a CCFRP aircraft is printed and the five different printing temperature control methods are applied to analyze the temperature difference. Compared with the spanner, the aircraft has many small angles between the straight lines. To utilize the fast printing advantages of 3DMP, the orange areas also have a printing speed of 20 mm/s, and the blue areas are 7 mm/s. As the measurement results of the spanner, the printing temperature controlled by fixed power follows the changing of printing speed, as shown in Fig. 4 (d). For the general PID and step-PID control methods, the standard deviation (SD) of temperature reduced compare with the fixed power control method. However, the measured maximum and minimum temperature are even worse than the fixed power control method, as shown in Fig. 4(e) and (f) and (i).

Because the ex-post PID control leads to that the high speed already switches to low speed, but the microwave power still maintains at high level and may even keep increasing. Therefore, the standard deviation, maximum and minimum printing temperature of prediction-model control methods are much lower than the fixed power, general PID and step-PID methods. Furthermore, by using the PMSP control method, the standard deviation reduces by 53% of fixed power, 56% of general PID and 51% of step-PID methods, as shown in Fig. 4(h) and (i).

Another complex CCFRP part with lots of curves and corners are shown in Fig. 5, and this biomorphic spider shape comes from the Nazca lines [19], and three different printing speed are applied, including 20 mm/s (orange areas), 13 mm/s (green areas) and 7 mm/s (blue areas), as illustrated in Fig. 5 (b). The fixed power control method for this complex spider part has a standard deviation of 44 °C, but the standard deviation (SD) of the PMSP temperature control method is only 18 °C, as

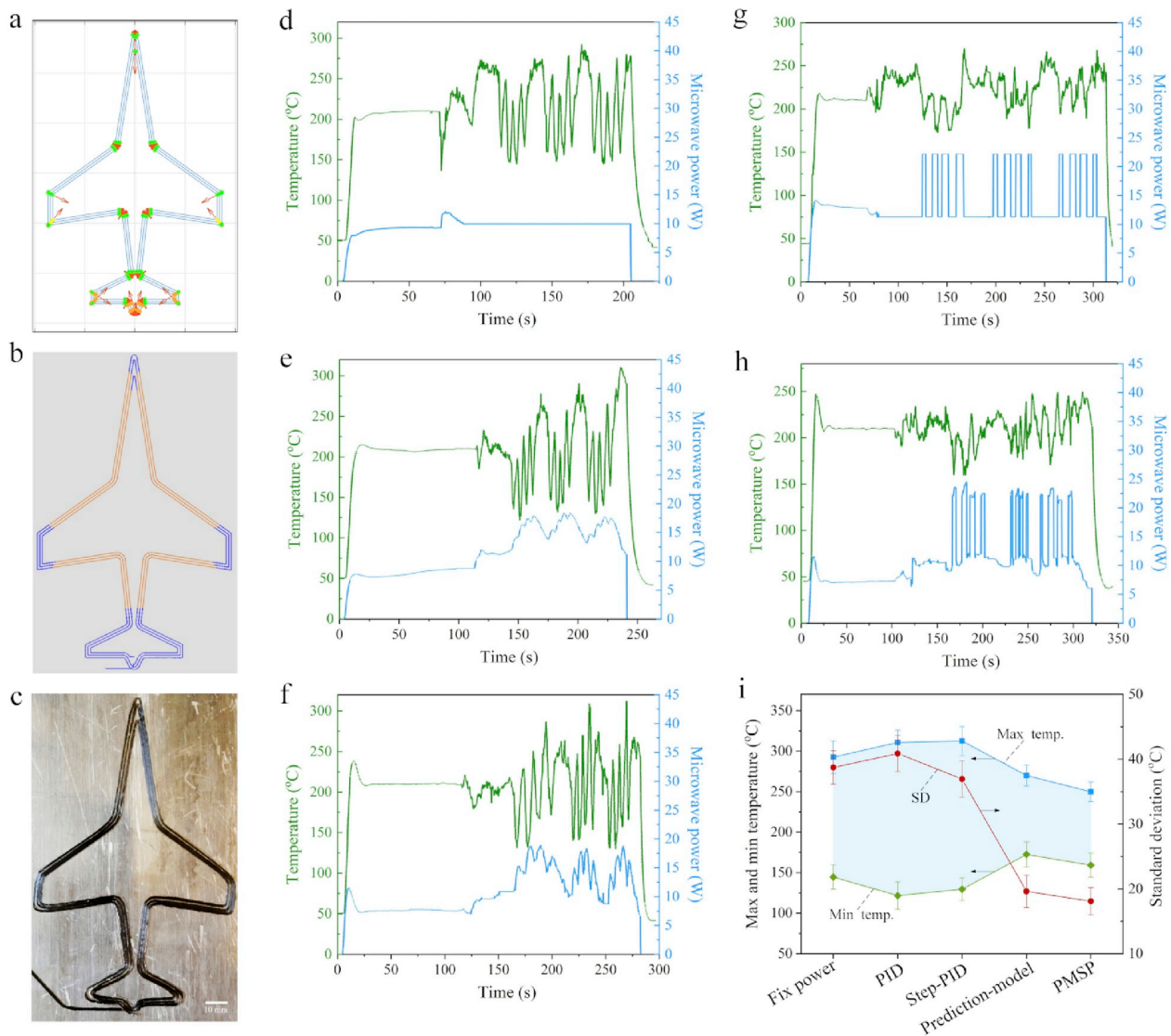


Fig. 4. 3D microwave printed CCFRP aircraft and measured printing temperature during the printing process, (a) calculated radius of curvatures of the printing path, (b) printing speed variation along the printing path of the spanner, (c) the printed CCFRP aircraft by PMSP method, (d) printing temperature under fixed microwave power, (e) printing temperature under general PID controlled microwave power, (f) printing temperature under step-PID controlled microwave power, (g) printing temperature under prediction-model controlled microwave power, (h) printing temperature under prediction-model and step-PID controlled microwave power, (i) the comparison of five different printing temperature control methods.

shown in Fig. 5 (d) and (h). Finally, the standard deviation further reduces by 53% and 34% compared with the step-PID and prediction-model based control method, by using the PMSP control method, as shown in Fig. 5(h) and (i). The CCFRP spider printed by the PMSP temperature control method is illustrated in Fig. 5 (c).

3.2. Optics inspection of 3D microwave printed CCFRP

As illustrated in Fig. 6, the 3D microwave printing technology can print different CCFRP shapes with variation printing speed. All of these optical figures have the same size bar shown in Fig. 6 (a). It is obvious that after the round filaments were heated by the microwave and printed on the platform, the filaments change the shape from cylinder to flat strip.

Consequently, the strip-filament has better bonding strength to each other and avoids the voids between them. From Fig. 6(a)–(c), the straight lines, curved lines and arc turning corner have the printing speed of 20 mm/s. Under a high printing speed, the filaments still can follow the printing path (yellow dash lines) and have very less

disordered carbon fibers, because of the rapid and volumetric microwave heating. Due to the strong stiffness of the carbon fiber bundle, the folding of the strip-filament on a sharp corner is expected as shown in Fig. 6(e) and (f). Even the direction vectors of the 180° circular corner and sharp angle corners are changing persistently, the filament can be printed, as shown in Fig. 6(d)–(f).

3.3. Mechanical performance of the 3D microwave printed CCFRP

To inspect the mechanical performance of the printed CCFRP, we measured tensile strength for a set of standard specimens printed by the 3DMP method at 10 and 20 mm/s speeds and compared them with those printed with conventional heating methods (the printing process information is described in the supplementary information, Fig. S6). As shown in Fig. 7 (a), with convectional printing nozzle that relies on contact heat transfer, the printing speed is only 5 mm/s (the printing nozzle is similar to the previous work [14]). And the corresponding tensile strength ranges from 253 MPa to 355 MPa (average value is 303 MPa). By applying the PMSP temperature control method, the tensile

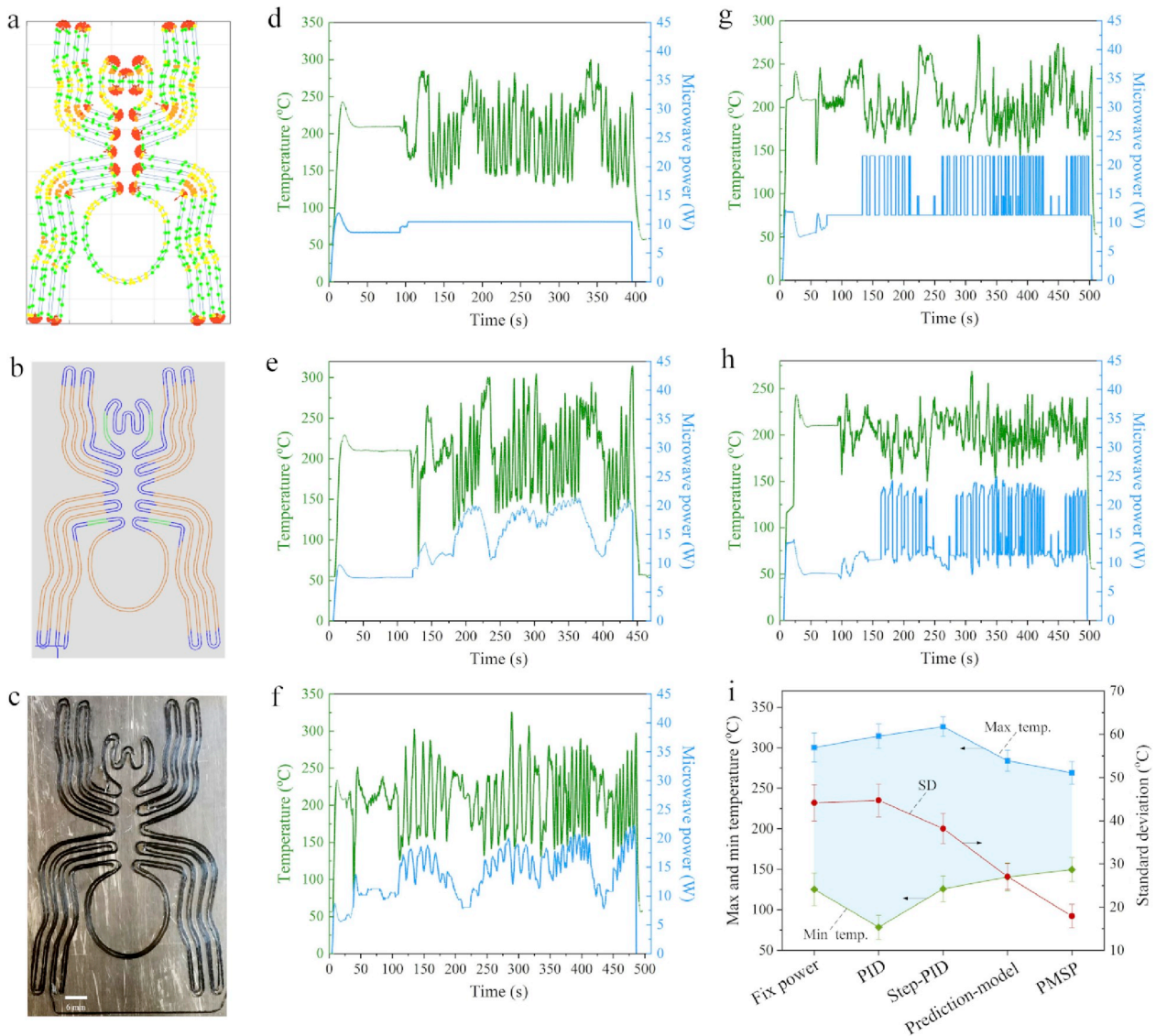


Fig. 5. 3D microwave printed CCFRP Nazca lines spider and measured printing temperature during the printing process, (a) calculated radius of curvatures of the printing path, (b) printing speed variation along the printing path of the spanner, (c) the printed CCFRP Nazca lines spider by PMSP method, (d) printing temperature under fixed microwave power, (e) printing temperature under general PID controlled microwave power, (f) printing temperature under step-PID controlled microwave power, (g) printing temperature under prediction-model controlled microwave power, (h) printing temperature under prediction-model and step-PID controlled microwave power, (i) the comparison of five different printing temperature control methods.

specimens were printed by using microwave under speed of 10 mm/s and 20 mm/s. At a higher speed of 20 mm/s, the tensile strength is measured to be between 287 and 411 MPa. The measured average tensile strengths are about 358 MPa for the two printing speed, as shown in Fig. 7 (a). These results clearly indicated that the mechanical performance of CCFRP fabricated by high speed 3DMP surpasses the one printed by conventional method. Additional mechanical data adapted from other researchers' works [6,7,12,20] that employed conventional printing methods are also plotted in Fig. 7 (a). It is indicated that the 3DMP has a superiority in both printing speed and mechanical performance. The micrographs of the specimens printed by 3DMP are illustrated in Fig. 7(b) and (c). The filament has been heated and pressed to the platform and eliminated the voids between each other. The higher heating temperature of carbon fiber than the matrix resin may bring better infiltration quality to the specimens.

4. Conclusions

The less efficient heat transfer of conventional heating leads to the

slow printing speed, small printing amount and low heating temperature. On the contrary, the 3D microwave printing technology provides the rapid, volumetric and selective heating advantages and can manufacture the high-performance continuous carbon fiber reinforced composites with short period. Because the curves or turning paths have the changing directional vectors during the printing process, the printing should be slow down to make sure the filament can adhere to the printing platform. Therefore, the variation printing speed is employed in the 3D microwave printing technology and a new printing temperature control method to reduce the heating temperature difference under variation printing speed is presented in this paper. A prediction-model and step-PID (PMSP) combined temperature control method has been presented and compared with other methods. The experiment results exhibited that the PMSP method dramatically reduces the standard deviation of the printing temperature to 66%, 56%, 60% of the CCFRP spanner, aircraft and Nazca lines spider respectively, by comparing with general PID control method. Moreover, the printed CCFRP at a speed of 20 mm/s has a tensile strength (average value of four specimens) up to 358 MPa, higher than the conventional printed one at a low speed of 5

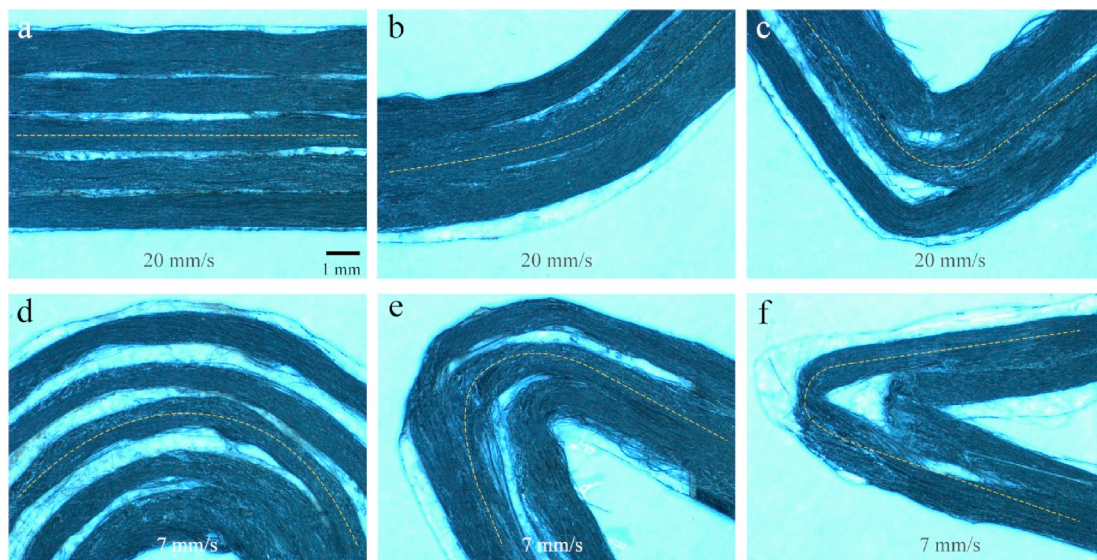


Fig. 6. Zoomed optical figures of 3D microwave printed CCFRP with different speed and shapes, (a) straight lines, (b) curved lines, (c) arc turning corner, (d) 180° circular corner, (e) 75° corner, (f) 23° corner.

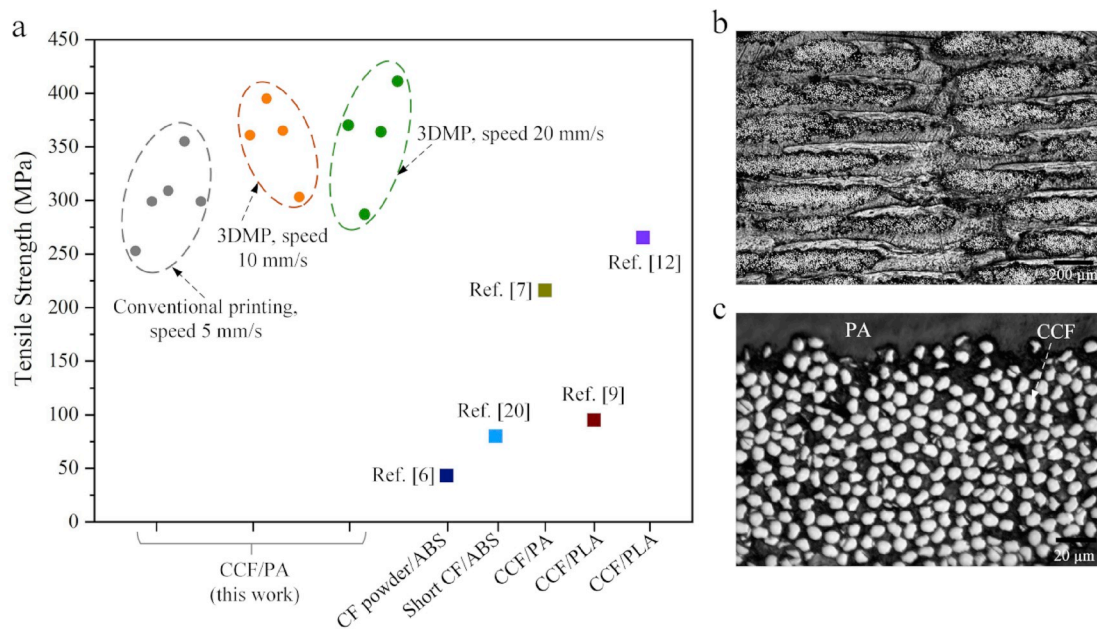


Fig. 7. Comparison of tensile strength between 3D microwave printing method (3DMP) with high speeds in this work and traditional printing methods, (a) tensile strength of printed composites, (b) and (c) micrographs of the specimens fabricated by 3DMP. CCF represents continuous carbon fiber, PA is polyamide, and PLA is polylactic acid. ABS represents acrylonitrile butadiene styrene.

mm/s. The ongoing research will use this technology to further investigate the printing speed and mechanical properties of the continuous carbon fiber reinforced polymer composites.

Author statement

Nanya Li: Conceptualization, Methodology, Software, Formal analysis, Writing - Original draft preparation. **Guido Link:** Investigation, Writing - Review & Editing. **John Jelonnek:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Acknowledgements

The authors would like to thanks the support of Alexander von Humboldt Foundation as well as to the donor, the German Federal Ministry for Education and Research. The authors also would like to thanks the programming by Mr. Ting Wang, and the help of mechanical test by Dr. Mario Walter and Mr. Ralf Dahm.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compscitech.2019.107939>.

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